Status	Finished
Started	Thursday, 15 October 2026, 9:05 PM
Completed	Thursday, 15 October 2026, 9:12 PM
Duration	7 mins 39 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Combining sorted sublists
- c. Comparing each element with the others
- d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **2**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Rearranging data without sorting
- c. Sorting data sequentially
- d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **3**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution
b. Selection
c. Extraction
d. Insertion

The correct answer is:
Insertion

Question **4**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It requires external libraries
b. It only works with integer arrays
c. It sorts data out of the box efficiently
d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Finding a specific element in a list
b. Removing duplicates from a list
c. Arranging data in a specific order
d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **6**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are ignored
b. Each part is sorted independently
c. Each part is searched for a specific element
d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **7**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Complexity
b. Sorting
c. Searching
d. Merging

The correct answer is:
Complexity

Question **8**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 200,470,80,150,40,90,300,400,70,120
b. 40,80,90,150,200,300,400,470,70,120
c. 40,70,80,90,120,150,200,300,400,470
d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **9**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are combined to form a single sorted list
c. They are compared element by element
d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **10**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes it easier to select items based on their relationship to the rest of the items
b. It slows down the process
c. It makes the data unsorted
d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Bubble Sort
b. Quick Sort
c. Merge Sort
d. Insertion Sort

The correct answer is: Bubble Sort

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Insertion Sort
b. Merge Sort
c. Quick Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It recursively sorts subproblems
b. It divides the list into sublists
c. It uses the heap data structure
d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting helps in finding duplicates quickly
b. Sorting decreases the efficiency of selection operations
c. Sorting makes it harder to search for items
d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Binary Search
b. Linear Search
c. Merge Sort
d. Bubble Sort

The correct answer is: Bubble Sort